

Using physics-informed AI to enable fibre TM recovery from minimal partial measurements enabling ultra-thin endoscopes

Callum J. Wilson^a, Yijie Zheng^b, David B. Phillips^c, and George S. D. Gordon^d

^{a, d}Optics and Photonics Research Group, University of Nottingham, United Kingdom

^bCavendish Laboratory, University of Cambridge, United Kingdom

^cDepartment of Physics and Astronomy, University of Exeter, United Kingdom

ABSTRACT

Multimode optical fibres offer a route to ultra-thin, high-resolution endoscopes but require frequent transmission matrix (TM) recalibration, which is too slow if the full TM must be measured experimentally. This work presents a physics-informed self-attention model that predicts the bend radius and direction of a multimode fibre from only a partial set of TM measurements, using 33% of the guided modes as input. The predicted bend parameters are passed to a physical forward model to reconstruct the full complex TM in the fibre eigenmode basis, enabling accurate TM recovery from compressed measurements. Simulation results show mean absolute bend-radius errors below 1% with corresponding TM errors approaching 0% for radii above 0.3m, demonstrating that near-complete TM information can be retrieved from sparse measurements while remaining consistent with the underlying fibre physics. This approach could substantially reduce calibration time and data requirements for flexible, ultrathin endoscopic imaging systems based on multimode fibres.

Keywords: Artificial intelligence, endoscopy, multimode fibre, transmission matrix recovery, deep learning, self-attention, physics informed

1. INTRODUCTION

Medical professionals use a wide range of tools to investigate a patient's symptoms and to check for signs of disease. One example is an endoscope which allows the user to look directly inside the human body. However traditional imaging systems use bulky lenses that restrict the lower limits of camera size¹ making imaging of deep tissues difficult. This causes problems as delays in detection and diagnosis of disease can affect outcomes. For example, early detection of pancreatic cancer can improve survival but requires imaging instruments small enough to fit in a needle ($< 0.7\text{mm}$).²

Multimode optical fibres have been proposed for use as cost-effective, precise,³ ultrathin ($< 0.1\text{mm}$ diameter) endoscopes. Multimode fibres (MMF) have larger core diameters than single-mode fibres but can guide multiple spatial modes of light simultaneously, enabling the formation of images as spatial information is captured. MMFs are roughly an order of magnitude smaller than traditional endoscopes and promise subcellular resolution imaging at the tip of a needle.⁴ Despite these advantages, MMFs suffer from image distortion due to unpredictable bending,⁵ movements, and temperature drifts;⁶ even small changes in configuration have been shown to ruin imaging abilities.⁷ If the fibre's transmission matrix (TM) is known, these distortions can be removed to produce clear images.⁸ The TM is a complex valued matrix that represents the relationship between input and output pixels, as long as the fibre is stable,⁹ see equation 1. $E_{\text{in}} \in \mathbb{C}^N$ and $E_{\text{out}} \in \mathbb{C}^N$ represent the complex-valued input and output optical fields, respectively, discretised and stacked into column vectors. The scalar N corresponds to the number of supported spatial modes in the fibre, which is equivalent to the number of independent spatial channels or resolvable pixels.¹⁰ TM has dimensions $N \times N$.

$$E_{\text{out}} = TM \cdot E_{\text{in}} \quad (1)$$

Further author information:

CJW e-mail: callum.wilson@nottingham.ac.uk

Previous work suggests that the effect of bending on the TM is predictable and could be used to infer the TM, but this is limited by the experimental lack of precise refractive index profile knowledge.¹¹ Furthermore, ambiguity can arise if multiple different bent states produce identical or near-identical TMs, and the number of unique perturbations is so large that exploring all possibilities becomes impractical.⁶ The TM is very sensitive to perturbations of the fibre, so up-to-the-minute characterisation of the TM is required for many applications,¹² such as real-time endoscopy. Perturbation correction must be done with access to only one end of the fibre too, the proximal end, which is located outside the body.⁶ The most advanced applications require knowledge of the full, spectrally resolved TM,¹³ extending acquisition time. The number of probe measurements needed to calibrate the TM can easily extend into the thousands.¹² Measuring the full TM experimentally can be too time-consuming for practical use, especially if high resolution images are required. Gao et al.¹⁴ explain how this is done in practice if both ends of the fibre are accessible. In short, known input fields, modulated by a device such as a spatial light modulator (SLM), are sent through the fibre, and the output is recorded by a camera. Phase and amplitude are recovered using digital holography or phase-shifting, and the TM is computed from these input-output pairs. The need for real-time performance for some tasks, specifically in endoscopic disease detection, diagnosis, and surgical procedures, is more critical,¹⁵ highlighting the need for efficient single-ended TM measurement in clinical translation.

A number of approaches have been proposed for recovering TMs using only one fibre end, though the problem remains unsolved in practice. Gordon et al.¹⁶ proposed using a stack of metasurface reflectors attached to the distal end of the fibre to determine the TM in a reflection-mode configuration (see Figure 1). Their algorithm used to recover the TM took several minutes to execute, which increases with the size of the TM. Zheng et al.⁵ built upon this work by training an AI algorithm to approximate the multi-step conventional reconstruction process with a substantially reduced execution time, showing that combining structured reflectors at three wavelengths and AI can recover the TM more quickly.

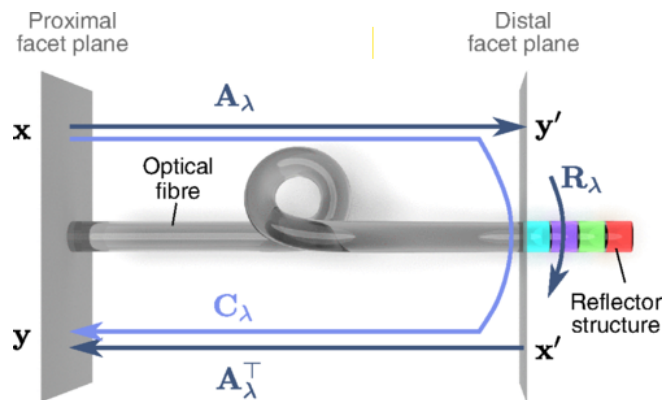


Figure 1. Metasurface stack applied to the distal end of a multimode optical fibre. A_λ represents the TM at wavelength λ .¹⁶

Wen et al.³ developed a system to recover the TM in milliseconds. Their work shows that under some limited bending conditions, a 'nearest neighbour' database look-up approach without a structured reflector (relying only on fibre facet Fresnel reflection) can work. Li et al.¹⁷ shows that a combination of structured reflectors/guidestars and prior knowledge can enable fast calibration. Together, these show the potential for combining reflections with prior knowledge of fibre structure and bending states to efficiently estimate the TM.

Similarly, it has been shown that the full TM can be sampled with only a subset of measurements (1-5%).¹² Our aim is therefore to develop a flexible AI model that exploits both prior knowledge of the fibre, which is encoded by physical modelling of the fibre as a sequence of bent segments, and a reduced measurement basis to flexibly exploit both of these and enable TM reconstruction without use of complex reflectors based on minimal measurements.

2. METHODS

The transmission matrix can be represented in various ways; the following work uses TMs in the fibre-eigenmode basis. The data for this project was collected using a simulation programme, using a core diameter of $8\text{ }\mu\text{m}$. The TMs were generated using the transfer matrix method (TMM), which includes modelling the fibre as a sequence of infinitesimally small segments, and using spatial translation between segments to simulate the mode coupling induced by bending.¹⁸ Initially, 8000 training TMs and 2000 test TMs were generated, covering single bend radii from -0.85 m to 0.85 m ; however bend radii between -0.05 m and 0.05 m were not considered. The radius for a given bend could be positive or negative, depending on the orientation of the bend. The TMs were comprised of 78×78 complex-valued numbers representing mode coupling. Most AI models don't natively handle complex-numbers, therefore the matrices were converted to 156×156 real number matrices (see equation 2). Each TM was paired with a single floating point number which represents the bending radius.

$$z = a + bi \quad \leftrightarrow \quad \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \quad (2)$$

2.1 AI model design

Various deep learning techniques were considered; including convolutional neural networks, self-attention, and multi-head attention. Upon comparison using a single train and validation split, single-head self attention performed best, consistent with previous work in this area.¹⁰ An initial attention model (see Figure 2) was developed to predict the bending radius of a multimode fibre based on its full transmission matrix. The model's mean absolute error was 1.33%.

This architecture was designed to progressively reduce the dimensionality of the TM (dense layers, pooling) while preserving long-range inter-modal dependencies (self-attention), which are expected due to strong mode coupling in multimode fibres. The ordering of layers reflects this design goal, moving from local feature compression to global relational modelling. The contribution of each component was validated through ablation analysis (Table 1), which informed the final architecture used in subsequent experiments and is described in detail in Section 2.2. Batch size and learning rate were kept constant throughout the experiment to ensure a fair comparison, while the number of epochs changed as early stopping was used to reduce overfitting.

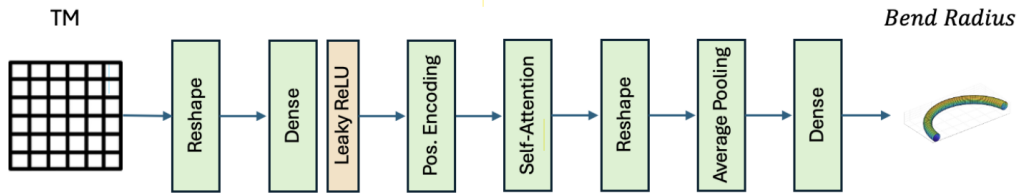


Figure 2. Initial attention-based model to predict fibre bend radii.

Table 1. Ablation results showing how each layer contributes to model performance.

Model	Error (%)	Training Time (s)	Batch Size	Epochs	Learning Rate	Parameters
Minus Positional Encoding Layer	0.67	41.8	31	37	0.00661	3665
Full Model	1.33	40.2	31	34	0.00661	4913
Minus Attention Residuals	1.77	26.5	31	23	0.00661	4881
Minus Global Pooling Layer	1.84	49.8	31	47	0.00661	7377
Minus Initial Dense Layer	1.94	35.3	31	27	0.00661	110761
Minus Attention Layer	17.40	65.9	31	100	0.00661	3793

2.2 Using partial TM measurements

Although the model could predict the bend radius using the full TM, a more practical approach is to use only a partial set of TM measurements and infer the rest. After predicting the bend radius from this reduced data, it can be passed to a forward physical model to estimate the full TM, reducing manual measurement effort and

potentially speeding up TM recovery for endoscopic imaging. To evaluate this idea, the model was retrained using only 33% of the fibre modes.

The simulation programme was altered so that only a subset of the available fibre modes were used. One third of the modes was chosen for initial experimentation as the full set of 78 fibre modes is evenly divisible by three, allowing for uniform downsampling by selecting every third mode. The programme calculated all available modes, and every third was kept and used to create the TMs. 12000 training and 3000 test TMs were generated.

The same architecture as Figure 2 was used to develop the model, minus the positional encoding layer (see Figure 3). TMs with 26 modes were created (represented as 52×52 real numbers) and batches of 28 TMs at a time were passed to the model for training. The batch size plus other hyperparameters (learning rate, layer embedding dimensions) were found using Bayesian optimisation hyperparameter tuning. Optimiser Adam was used with a learning rate of 0.005565, with a layer embedding dimension of 22.

Firstly, each TM was reshaped into 26, 2×52 patches, each representing one set of mode measurements. This idea is similar to a vision transformer splitting an image into patches, reducing computational load. Next each patch was passed to a dense/fully connected layer with a hyperbolic tangent activation function. This layer reduced the dimensions of each patch independently to a latent space and introduced non-linearity. This new representation of the data contains only relevant information. Layer normalisation was used to provide stability and improves generalisation.

The data was then passed to the main self-attention layer. Self-attention, a technique derived from transformer architectures,¹⁹ enables the model to capture long-range dependencies without assuming local correlations. This contrasts with 2D convolutional neural networks, which rely on spatial locality — an assumption that is valid for image data but not for transmission matrices, where spatial structure is lost when 2D fields are vectorised and further nonlocality may be introduced by a change of basis, such as the mode basis used here. The self attention layer also contained residual connections, where the input to the layer is added to the output. This provides stability by ensuring the model gradually updates during training. The self-attention output is reshaped in preparation for the global average pooling layer which only accepts three dimensional data. This layer summarises the data by averaging across every position of all patches, in turn helping the model generalize better and reduce overfitting. The final dense layer has one neuron in its output, which corresponds to the predicted bend radius. It has no activation function, so acts as a purely linear transformation.

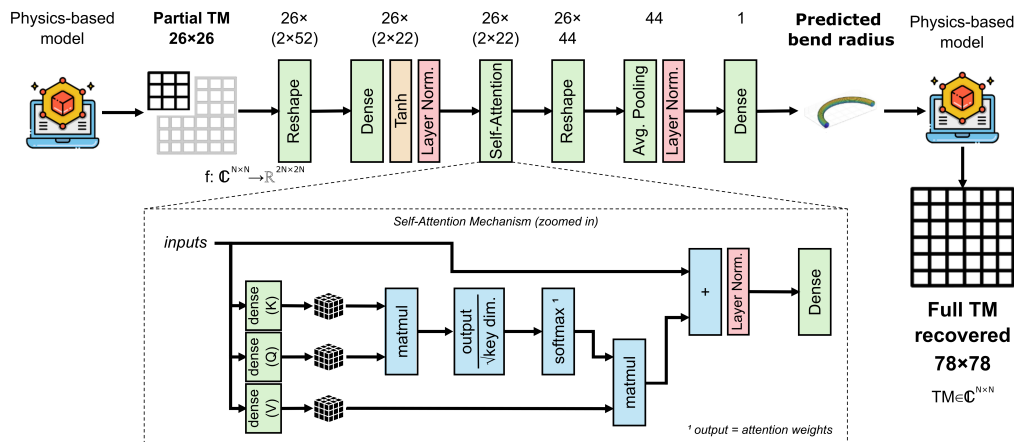


Figure 3. Attention-based model that predicts fibre bend radius from a partially measured transmission matrix and reconstructs the full transmission matrix via a forward physical model, forming a cycle-consistent workflow.

A custom loss function (equation 3) was designed to guide the model during backpropagation. It first computes the absolute percentage error for each prediction, which is then multiplied by a weight. This weight is the reciprocal of the true value plus a small constant, encouraging the model to prioritise minimising errors for smaller bend radii. The second term in the equation is a balance penalty, which penalises the model if there is a large discrepancy between prediction errors for positive and negative bend directions.

$$\text{Loss} = \left(\frac{1}{N} \sum_{i=1}^N \left(\frac{1}{|y_{\text{true},i}| + x} \cdot \frac{|\text{error}_i|}{|y_{\text{true},i}|} \right) \right) \cdot (1 + |\text{MAE}_{\text{pos}} - \text{MAE}_{\text{neg}}|) \quad (3)$$

Where N is the number of samples; $y_{\text{true},i}$ is the true value for sample i ; $\text{error}_i = y_{\text{true},i} - y_{\text{pred},i}$; x is a small constant added for numerical stability (depends on the sign of $y_{\text{true},i}$). If $y_{\text{true}} < 0$, then $x = 0.01$; if $y_{\text{true}} > 0$, then $x = 0.004$, found using hyperparameter tuning; MAE_{pos} is the mean absolute error for samples where $y_{\text{true}} > 0$, computed over the current batch; MAE_{neg} is the mean absolute error for samples where $y_{\text{true}} < 0$, computed over the current batch.

3. RESULTS

Figure 4 (left) shows predicted bend radii vs. the ground truths. The aim was for points to be as close to the identity line as possible. The model was able to predict bend radii and direction, indicating that the direction of a bend does in fact alter the transmission matrix. Performance drops slightly for bend radii > 0.6 m, this is likely due to the loss function aggressively forcing the model to concentrate on smaller radii.

The trained model was now evaluated by predicting the bend radii when provided with TMs for bends: -0.7015 m, -0.3261 m, -0.084 m, 0.084 m, 0.3261 m, 0.7015 m. The aim was to calculate the percentage errors for each radius (as seen on the primary y-axis of Figure 4, right), and to recover the full 78×78 complex TM for each bend. Each predicted bend radius was input to the initial simulation programme. For example, for bend radius 0.3261 m, the predicted value was 0.32676 m, producing a percentage error of 0.202%. Both 0.3261 m and 0.32676 m were given as inputs to the simulation programme and the full TMs for each bend were saved so they could be compared; the idea being that if the two TMs are the same or very similar, it provides evidence that only a partial TM needs measuring to recover the full matrix.

The two TMs were compared by computing the phase aligned error, referred to as 'TM error'. Phase error finds the overall phase difference between the two matrices. TM error (equation 4) measures how different the predicted TM is from the true TM after correcting for this phase difference. A smaller percentage indicates that the two TMs are closer.

$$\text{TM error (\%)} = \frac{\|B - A e^{-j(\arg(\text{tr}(AB^H)))}\|}{\|A\|} \times 100 \quad (4)$$

Figure 4 (right; secondary y-axis) shows TM errors vs. radii tested. An exponential trend was observed: TM error is higher for small bend radii and decreases as radii increase, indicating that TM errors aren't consistent across all bend radii. The reduced prediction accuracy observed at small bend radii may be partly attributable to increased instability in the simulated TMs. For tight bends, the discretised simulation approach can introduce segment-to-segment mismatch, leading to reduced modal overlap and increased sensitivity to numerical perturbations. This results in noisier TMs, which may be more difficult for the model to learn from compared to the smoother, more stable TMs obtained at larger bend radii. This behaviour motivated the use of a weighted loss penalty, emphasising accuracy at low bend radii. For fibres that have bend radii > 0.3 m, the TM can be recovered with almost 0% error using only one third of the available fibre modes.

4. DISCUSSION

This work demonstrates that a multimode fibre transmission matrix can be recovered using a partial set of modes, potential improving acquisition and computation speeds. This was achieved by creating a physics based model that compresses the fibre to a physically interpretable 'bend' state and from this uses a forward physical model to reconstruct the TM. So far 26 modes can be used to recover a 78 mode transmission matrix accurately, along with other models requiring less modes.

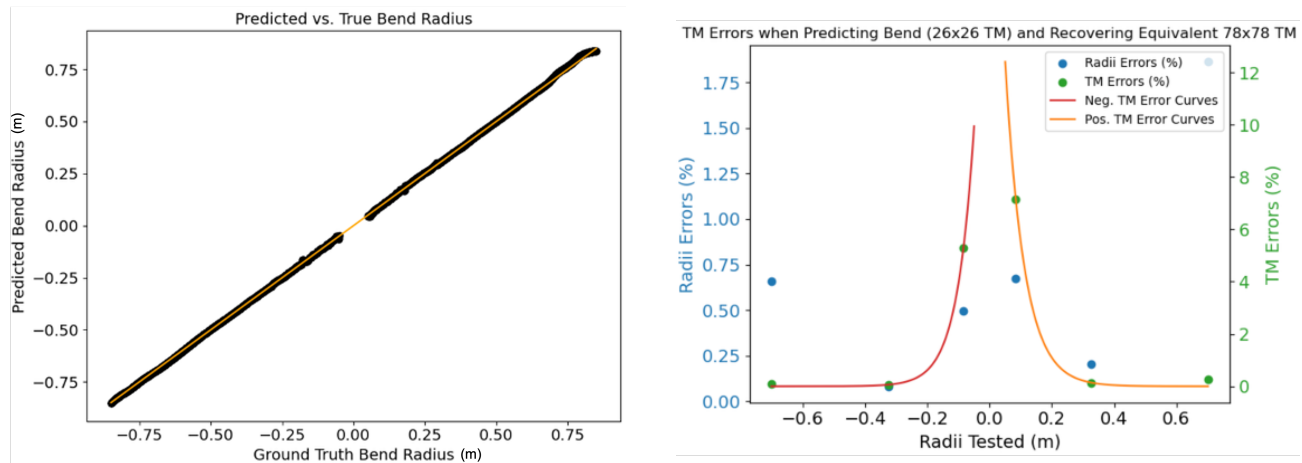


Figure 4. Absolute radii errors (left) and TM errors (right) when using 33% of available modes.

Future work will look to predict multiple bends simultaneously so transmission matrices of complex fibre shapes can be recovered too.

ACKNOWLEDGMENTS

CJW was supported by the Engineering and Physical Sciences Research Council (UKRI) through a doctoral studentship. GSDG would like to acknowledge his UKRI Future Leaders Fellowship (MR/Y034163/1 and MR/T041951/1).

CODE AVAILABILITY

The code used to train the AI models is available upon request via a private GitHub repository.

REFERENCES

- [1] Tseng, E., Colburn, S., Whitehead, J., Huang, L. C., Baek, S. H., Majumdar, A., and Heide, F., “Neural nano-optics for high-quality thin lens imaging,” *Nature Communications* **12**(1), 6493 (2021). Accessed: May 20, 2025.
- [2] Zhang, L., Pleskow, D. K., Turzhitsky, V., Yee, E. U., Berzin, T. M., Sawhney, M., Shinagare, S., Vitkin, E., Zakharov, Y., Khan, U., Wang, F., Goldsmith, J. D., Goldberg, S., Chuttani, R., Itzkan, I., Qiu, L., and Perelman, L. T., “Light scattering spectroscopy identifies the malignant potential of pancreatic cysts during endoscopy,” *Nature Biomedical Engineering* **1**(4) (2017). Accessed: December 16, 2025.
- [3] Wen, Z., Dong, Z., Deng, Q., Pang, C., Kaminski, C. F., Xu, X., Yan, H., Wang, L., Liu, S., Tang, J., Chen, W., Liu, X., and Yang, Q., “Single multimode fibre for in vivo light-field-encoded endoscopic imaging,” *Nature Photonics* **17**(8), 679–687 (2023). Accessed: May 30, 2025.
- [4] Phillips, D. B. and Faccio, D., “Section 10: Imaging through multimode fibers,” *Journal of Optics* **24**(8), 23–25 (2022). Part of: Ilaria Cristiani et al 2022 J. Opt. 24 083001. Accessed: May 20, 2025.
- [5] Zheng, Y., Wright, T., Wen, Z., Yang, Q., and Gordon, G. S. D., “Single-ended recovery of optical fiber transmission matrices using neural networks,” *Communications Physics* **6**(1), 306 (2023). Accessed: May 20, 2025.
- [6] Gordon, G. S. D., “Flexible ultra-thin super-resolution endoscopy,” *Science China - Physics Mechanics & Astronomy* **67**(3), 234232 (2024). Accessed: May 30, 2025.
- [7] Amitonova, L. V., “Multimode fiber endoscopes for computational brain imaging,” *Neurophotonics* **11**(1), S11509 (2024). Accessed: May 20, 2025.
- [8] Kim, M., Choi, W., Choi, Y., Yoon, C., and Choi, W., “Transmission matrix of a scattering medium and its applications in biophotonics,” *Optics Express* **23**(10), 12648–12668 (2015). Accessed: May 20, 2025.

- [9] Popoff, S., Lerosey, G., Fink, M., Boccard, A. C., and Gigan, S., “Image transmission through an opaque material,” *Nature Communications* **1**(81) (2010). Accessed: May 23, 2025.
- [10] Zheng, Y., Kilpatrick, R. J., Phillips, D. B., and Gordon, G. S. D., “Self-attention-based non-linear basis transformations for compact latent space modelling of dynamic optical fibre transmission matrices,” (2024). Accessed: May 23, 2025.
- [11] Flaes, D. E. B., Stopka, J., Turtaev, S., de Boer, J. F., Tyc, T., and Čizmar, T., “Robustness of light-transport processes to bending deformations in graded-index multimode waveguides,” *Physical Review Letters* **120**(23), 233901 (2018). Accessed: May 20, 2025.
- [12] Li, S., Saunders, C., Lum, D. J., Murray-Bruce, J., Goyal, V. K., Čizmar, T., and Phillips, D. B., “Compressively sampling the optical transmission matrix of a multimode fibre,” *Light: Science & Applications* **10**(88) (2021). Accessed: May 23, 2025.
- [13] Lee, S.-Y., Parot, V. J., Bouma, B. E., and Villiger, M., “Efficient dispersion modeling in optical multimode fiber,” *Light: Science & Applications* **12**(31) (2023). Accessed: May 23, 2025.
- [14] Gao, H., Hu, H., Zhang, Y., Zhang, W., and Yan, T., “Seeing through multimode fibers with physics-assisted deep learning,” *Optics and Laser Technology* **167**, 109761 (2023). Accessed: Jun. 02, 2025.
- [15] Ali, S., “Where do we stand in ai for endoscopic image analysis? deciphering gaps and future directions,” *npj Digital Medicine* **5**(184) (2022). Accessed: May 23, 2025.
- [16] Gordon, G. S. D., Gataric, M., Ramos, A. G. C. P., Moushah, R., Williams, C., Yoon, J., Wilkinson, T. D., and Bohndiek, S. E., “Characterizing optical fiber transmission matrices using metasurface reflector stacks for lensless imaging without distal access,” *Physical Review X* **9**(4), 041050 (2019). Accessed: May 30, 2025.
- [17] Li, S., Horsley, S. A. R., Tyc, T., Čizmar, T., and Phillips, D. B., “Memory effect assisted imaging through multimode optical fibres,” *Nature Communications* **12**(1), 3751 (2021). Accessed: May 30, 2025.
- [18] Argyros, A., Lwin, R., and Large, M. C. J., “Bend loss in highly multimode fibres,” *Optics Express* **16**(23), 18590–18598 (2008). Accessed: Jun. 16, 2025.
- [19] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I., “Attention is all you need,” in *Advances in Neural Information Processing Systems*, **30**, 5998–6008 (2017).